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Vehicle cam jack

Abstract

A jack comprises a planar frame having a support post and an arcuate edge member. A pivot socket is provided at a first end of the support post that is sized to receive and engage an axle portion of a vehicle. The arcuate edge member has a first end extending along the pivot socket, a second end adjacent a second end of the support post, and an outer arcuate edge extending outwardly between the first and second ends of the support post. The outer arcuate edge defines a radius of curvature greater than a length dimension of the support post. Thus, the jack is positionable between a first orientation to allow the pivot socket to receive and engage an axle of a vehicle and a second orientation to allow the support post to lift the axle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(2) Not Applicable.

FIELD OF INVENTION

(3) The present general inventive concept pertains to vehicle jacks, and more particularly to a vehicle cam jack designed to safely and conveniently elevate and lower a portion of a vehicle using forward and reverse motion of the vehicle.

BACKGROUND

(4) In the use of wheeled vehicles, such as automobiles, trailers, etc., it occasionally becomes necessary to raise and lower one or more wheels of the vehicle in order to gain more convenient access to the wheel or an underside portion of the vehicle. For example, in the case of vehicles employing compressed air-filled tires, repair of a flat tire may require that the vehicle, or a portion thereof, be lifted in order to raise the tire from contact with the ground. The raised tire may then be removed from the vehicle and replaced or repaired.

(5) Numerous devices exist to accomplish raising and lowering of a vehicle or a portion thereof, such devices commonly referred to as “jacks.” One type of jack is the so-called “cam” jack, which, in various designs, uses an arcuately shaped support member having a pivot attachment at one end which is adapted to engage an axle of the vehicle. The arcuate support member defines an arcuate rolling surface, such as for example a spiral shape or a series of segmented surfaces defining an oblong shape, such that when the axle of the vehicle is engaged by the pivot attachment at one end of the cam jack and the vehicle is rolled toward the opposite end of the cam jack, rotation of the cam jack beneath the vehicle results in the cam jack lifting the engaged axle and an associated portion of the vehicle from the vehicle's resting configuration. In typical designs, a cam jack is sized to correspond to a particular height, such that the cam jack may be easily placed beneath a vehicle to engage an axle of the vehicle, and such that the lifting provided by the cam jack is sufficient to lift at least one wheel of the vehicle from the ground.

(6) Cam jacks often provide certain advantages over other designs of jacking devices. For example, in many designs, cam jacks are composed of a single unitary piece, rather than a series of interconnected moving parts. Hence, a cam jack often exhibits increased durability and a decreased chance of mechanical malfunction or misuse compared to other more complex mechanical jacking devices. Additionally, because cam jacks may consist of a single piece, they are often able to be deployed more rapidly and with less effort on the part of the user than other jack designs. In various designs, cam jacks are designed to occupy a relatively flat, planar shape. Thus, many cam jacks may be stored conveniently within a vehicle or adjacent a vehicle's exterior. However, though many designs of cam jacks offer numerous advantages over other mechanical jack designs, several cam jack designs also suffer from several shortcomings.

(7) For example, in many designs, cam jacks are provided with pivot attachments that fit loosely around the axle of a vehicle and/or that do not retain the axle within the socket. In such designs, particular care must be paid that the cam jack is installed securely against the axle, along a near perfectly vertical plane that extends perpendicular to a rolling direction of the axle. Otherwise, misalignment of the cam jack in relation to the axle and/or a vertical orientation to the ground may result in the cam jack “slipping out” from engagement with the axle, either by the cam jack becoming disengaged from the axle itself, or by the cam jack sliding along the ground surface due to non-vertical loads placed on the cam jack by the vehicle. This may be particularly problematic in cam jacks that are used on uneven ground or on an inclined surface.

(8) Additional shortcomings may be encountered in a cam jack in which no feature is provided to prevent over-rotation of the cam jack once the vehicle is rolled to a “lifted” orientation of the jack. In many cam jack designs, allowing the vehicle to roll a limited distance and allowing the cam jack to rotate beneath the vehicle to a certain degree may result in the cam jack lifting the axle from a resting position. However, continued rolling of the vehicle and rotation of the cam jack beyond the optimal distance/degree for lifting of the vehicle may result in “over-rotation,” wherein the cam jack effectively “tips over” beyond the lifted orientation and is no longer positioned fully beneath the axle to support the vehicle in the elevated position. In certain situations, excessive over-rotation of the cam jack may result in the vehicle being re-lowered to the ground, and in other more extreme situations, such over-rotation may result in uncontrolled dropping of the vehicle, creating a potentially unsafe condition for objects and individuals in the vicinity of the jack and/or the vehicle.

(9) In light of the above, there exists a desire to provide an improved vehicle cam jack which addresses one or more of the above-discussed limitations while remaining relatively quick and easy to store and deploy to lift a wheel and/or axle portion of a vehicle. There is a desire to provide an improved vehicle cam jack which may, in various embodiments, limit or control the negative effects of over-rotation of the cam jack on maintaining lift of the vehicle. There is a further desire to provide an improved vehicle cam jack which may, in various embodiments, provide a more safe and secure lifting mechanism for lifting a vehicle than devices currently known in the prior art, or which may provide a lifting mechanism which may be used more evenly and more conveniently on uneven or non-horizontal terrain, as compared to other devices currently known in the prior art.

BRIEF SUMMARY OF THE INVENTION

(10) According to various example embodiments of the present general inventive concept, a jack for lifting an axle portion of a vehicle is provided that includes generally a planar frame comprising a support post and an arcuate edge member, and a pivot socket defined at a first end of the support post. Various example embodiments of the present general inventive concept may include a pivot socket which is sized to receive and engage an axle portion of a vehicle. Various example embodiments of the present general inventive concept may include an arcuate edge member that has a first end extending along the pivot socket, a second end adjacent a second end of the support post, and an outer arcuate edge extending outwardly between the first and second ends of the support post. Various example embodiments of the present general inventive concept may include an outer arcuate edge which defines a radius of curvature greater than a length dimension of the support post. In various example embodiments of the present general inventive concept, the jack may be positionable in a first orientation with the support post extending horizontally and the outer arcuate edge below the support post to allow the pivot socket to receive and engage an axle of a vehicle. In various example embodiments, the jack may be rotatable along the outer arcuate edge to a second orientation to allow the support post to extend beneath the pivot socket and rest on a portion of the outer arcuate edge adjacent the second end of the support post.

(11) The foregoing and/or other aspects and advantages of the present general inventive concept may be achieved by providing a jack in which the first end of the support jack forms a first side of the pivot socket. Additional aspects and advantages of the present general inventive concept may be achieved by providing a jack in which the first end of the support jack defines a substantially straight edge. In various embodiments, the jack may further comprise a contact plate extending perpendicularly along the straight edge of the support jack first end. In various embodiments, the frame may further comprise a plurality of bracing members extending between the support post and the arcuate edge member. In various embodiments, a first of the bracing members may extend from the first end of the support post to a portion of the arcuate edge member proximate the first end of the arcuate edge member, the first bracing member forming a second side of the pivot socket. In various embodiments, the first end of the arcuate edge member may form a third side of the pivot socket. In various embodiments, the second end of the support post may define a substantially straight edge extending acute to a long dimension of the support post.

(12) In various embodiments, the jack may further comprise a stabilizing plate extending perpendicularly along at least a portion of the outer arcuate edge of the arcuate edge member. In various embodiments, the outer arcuate edge and the stabilizing plate may each extend along the substantially straight edge of the support post second end to define a support platform. In various embodiments, the stabilizing plate may further extend along the outer arcuate edge between a midpoint of the outer arcuate edge and the support post second end. In various embodiments, the stabilizing plate may be tapered outwardly from midpoint of the outer arcuate edge to the support post second end. In various embodiments, a plurality of cleats may be defined along the outer arcuate edge at spaced apart locations between the midpoint of the outer arcuate edge and the support post second end. In various embodiments, each of the bracing members may extend from the support post first end to one of a plurality of spaced apart locations along the arcuate edge

member. In various embodiments, the support post, bracing members, arcuate edge member, and cleats may be integrally formed. In various embodiments, the support post, bracing members, arcuate edge member, and cleats may be formed from a unitary sheet of rigid material.

(13) Additional aspects and advantages of the present general inventive concept will be set forth in part in the description, drawings, and claims which follow, and, in part, will be obvious from the description, or may be learned by practice of the present general inventive concept.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The following example embodiments are representative of example techniques and structures designed to carry out the objects of the present general inventive concept, but the present general inventive concept is not limited to these example embodiments. In the accompanying drawings and illustrations, the sizes and relative sizes, shapes, and qualities of lines, entities, and regions may be exaggerated for clarity. A wide variety of additional embodiments will be more readily understood and appreciated through the following detailed description of the example embodiments, with reference to the accompanying drawings in which:

(2) FIG. 1 illustrates a first perspective view of a vehicle cam jack constructed according to one example embodiment of the present general inventive concept;

(3) FIG. 2 illustrates a second perspective view of the vehicle cam jack of the example embodiment illustrated in FIG. 1;

(4) FIG. 3 illustrates a side elevation view of the vehicle cam jack of the example embodiment illustrated in FIG. 1;

(5) FIG. 4 illustrates a side schematic view of a wheel of a vehicle and a corresponding axle, showing the vehicle cam jack of the example embodiment illustrated in FIG. 1 positioned on the axle in a first position; and

(6) FIG. 5 illustrates a side schematic view of the vehicle wheel of FIG. 4, showing the vehicle cam jack of the example embodiment illustrated in FIG. 1 positioned on the axle in a second position.

DETAILED DESCRIPTION

(7) Reference will now be made to the example embodiments of the present general inventive concept, examples of which are illustrated in the accompanying drawings and illustrations. The example embodiments are described herein in order to explain the present general inventive concept by referring to the figures.

(8) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the structures and fabrication techniques described herein. Accordingly, various changes, modification, and equivalents of the structures and fabrication techniques described herein will be suggested to those of ordinary skill in the art. The progression of fabrication operations described are merely examples, however, and the sequence type of operations is not limited to that set forth herein and may be changed as is known in the art, with the exception of operations necessarily occurring in a certain order. Also, description of well-known functions and constructions may be simplified and/or omitted for increased clarity and conciseness.

(9) Note that spatially relative terms, such as “up,” “down,” “right,” “left,” “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over or rotated, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The device may be otherwise

oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(10) According to various example embodiments of the present general inventive concept, a vehicle cam jack, or “jack,” is provided that includes a substantially rigid support frame having an arcuate outer edge extending along an outwardly curved path along a perimeter thereof between opposite first and second ends. The jack includes a pivot socket defined at a first end of the frame and a substantially flat support surface defined at an opposite second end of the frame. The pivot socket is sized, shaped, and configured to receive an axle of a vehicle rotatably therein and to maintain the axle in an orientation perpendicular to the arcuate outer edge of the frame. The support surface is positioned to extend along a plane substantially parallel to, and spaced apart from, an axial dimension of a vehicle axle received within the socket, and substantially perpendicular to the arcuate outer edge of the frame. The radius of curvature of the arcuate outer edge is generally shorter than the distance between the socket, at one end of the frame, and the support surface, at the opposite end of the frame. Thus, when the jack is placed in a first orientation, with the socket and the support surface in a substantially horizontal side-by-side relationship and the arcuate outer edge extending generally downwardly therebetween, the jack may be positioned beneath a vehicle and a vehicle axle may be received within the socket. Thereafter, the jack may be rolled along the arcuate outer edge toward a second orientation, in which the support surface is oriented substantially beneath the socket, thus allowing the greater distance between the socket and the support surface to force separation between the vehicle axle and the surface underlying the jack.

(11) FIG. 1 illustrates a jack **10** constructed in accordance with one example embodiment of the present general inventive concept. In the example embodiment illustrated in FIG. 1, the jack **10** includes a support frame **12** which is fabricated from a unitary, substantially planar sheet of rigid material, such as steel, iron, polymer, or the like, and has a plurality of central through openings **14** defined therein. The through openings **14** are provided to reduce the overall weight of the jack **10**, and the through openings **14** cooperate to define a plurality of integrally-formed support members extending along a planar orientation between the various through openings **14** in the support frame **12**. More specifically, the support frame **12** in the illustrated embodiment defines generally an elongated, flat, substantially straight support post **16** having a first end **18** and opposite second end **20**. As will be discussed in additional detail below, a pivot socket **22** is defined at a first end **18** of the support post **16**. The support frame **12** further defines a flat, arcuate edge member **24** extending along an outwardly curved path coplanar with the support post **16** from the second end **20** of the support post **16** to a distal end of the pivot socket **22** opposite from the support post first end **18**. The support frame **12** further defines a plurality of elongated bracing members **26**, **28**, **30**, with each bracing member extending from a respective location proximate the support post first end **18** to a respective location along the arcuate edge member **24**.

(12) With reference to FIGS. 1-3, in the illustrated embodiment, the pivot socket **22** is defined by certain components of the frame **12** proximate the support post first end **18**, as well as a contact plate **32** extending along an upper edge of the support post **18**, perpendicular to the plane of the frame **12**. More specifically, in the illustrated embodiment, the support post first end **18** defines a substantially straight upper edge **34** extending generally perpendicular to a long dimension of the support post **16**. A contact plate **32** is provided having a generally flat, rectangular shape with a long dimension corresponding to a length of the upper edge **34** of the support post first end **18**. The contact plate **32** is fixed along the upper edge **34** of the support post **16** as by a suitable weld, adhesive, integral connection, or the like, and the contact plate **32** extends in a plane generally parallel to the upper edge **34** of the support post **16** and perpendicular to the horizontal plane defined by the support frame **12**.

(13) In the illustrated embodiment, a first of the bracing members **26** extends inwardly from an inward side of the support post first end **18** and curves distally away from the support post first end **18** along a direction generally parallel with the long dimension of the support post **16**. Thus, the

first bracing member **26** extends from a region of the support post first end **18** adjacent the contact plate **32** to a region proximate a first end **36** of the arcuate edge member **24** which overhangs the first end **18** of the support post **16**. In this configuration, respective inner edges of the first end **36** of the arcuate edge member **24** and the first bracing member **26** cooperate with the contact plate **32** to form a spanner shaped pivot socket **22** having an open side **38** facing outwardly from the support post first end **18**, in a direction parallel to the plane of the contact plate **32**, parallel to the plane of the frame **12**, and perpendicular to the long dimension of the support post **16**.

(14) With reference to FIGS. **1** and **2**, in various embodiments, a stabilizing plate **50** is provided extending along a portion of the outer arcuate edge of the arcuate edge member **24**. In the illustrated embodiment, the stabilizing plate **50** consists of a rigid, arcuate plate conforming along the outer arcuate edge of the arcuate edge member **24** and extending generally outwardly from either side of the arcuate edge member **24**, perpendicular to the frame **12**. In the present embodiment, the stabilizing plate **50** defines a tapered shape, beginning at an approximate midpoint of the outer arcuate edge of the arcuate edge member **24** and widening on either side of the arcuate edge member **24** toward an end of the stabilizing plate **50** adjacent the second end **20** of the support member **16**. In the illustrated embodiment, a second end **54** of the arcuate edge member **24** adjacent the second end **20** of the support post **16** defines a substantially straight portion of the outer arcuate edge. Those portions of the stabilizing plate **50** adjacent the second end **54** the arcuate edge member **24** are correspondingly flat in order to conform to the straight portion second end **54** of the arcuate edge member **24** and to extend orthogonally along the outer arcuate edge. Thus, the portions of the stabilizing plate **50** and the second end **54** of the arcuate edge member **24** adjacent the second end **20** of the support member **16** cooperate to define a relatively flat, rigid support platform **56** upon which the support post **16** may rest. In the illustrated embodiment, the support platform **56** extends along the second end **54** of the arcuate edge member **24** at an angle slightly acute to an inside surface of the support post **16** facing the arcuate edge member **24**. Thus, when the support post **16** rests upon the platform **56** with the platform substantially flat on a horizontal surface, the support post **16** is biased slightly toward the arcuate edge member **24**.

(15) In various embodiments, a plurality of cleats **52** are provided at spaced apart locations along the outer arcuate edge of the arcuate edge member **24** in order to increase traction along the outer arcuate edge of the arcuate edge member **24**. In the illustrated embodiment, the cleats **52** are disposed at spaced apart locations between the approximate midpoint of the outer arcuate edge of the arcuate edge member **24** and the platform portion **56** of the stabilizing plate **50** adjacent the second end **20** of the support member **16**. However, it will be recognized that additional cleats may be provided at other locations along the outer arcuate edge of the arcuate edge member **24** without departing from the spirit and scope of the present general inventive concept. Furthermore, it will be recognized that numerous other devices, shapes, and configurations exist to provide increased traction along the outer arcuate edge of the arcuate edge member **24**, and such alternate devices, shapes, and configurations may be used without departing from the spirit and scope of the present general inventive concept.

(16) With reference to FIGS. **3** and **4**, it will be recognized that the jack **10** may be brought to a first orientation **40**, in which the open side **38** of the pivot socket **22** faces generally upward, the support post **16** extends generally horizontally, and the arcuate edge member **24** extends beneath the support post **16** between the first and second ends **18**, **20** thereof. In this first orientation **40**, the jack **10** may be placed beneath a vehicle having an axle **42**, and the axle **42** may be received within the pivot socket **22**, with a central portion of the arcuate edge member **24** resting on the ground beneath the vehicle. Thereafter, the jack may be moved to a second orientation **44** (see FIG. **5**), in which the frame **12** is rotated about the pivot socket **22** such that the contact plate **32** is substantially below the axle **42**, the open side **38** of the pivot socket **22** faces generally horizontally from the axle **42**, and the support post **16** extends generally downwardly between the axle **42** and a portion of the ground beneath the vehicle. In this second orientation, the platform **56** rests upon the

ground beneath the vehicle and the support post **16** rests upon the platform **56**, with the support post **16** extending at a slight angle to vertical toward the arcuate edge member **24**. Thus, when the jack **10** is brought to the first orientation **40** and placed beneath a vehicle in engagement with an axle **42**, the narrow end of the support plate **50** is positioned at a lower-most point of the arcuate edge member **24** nearest the ground. As the jack **10** is rotated toward the second orientation **44**, the portion of the stabilizing plate **50** contacting the ground beneath the vehicle changes, favoring the wider end of the stabilizing plate **50**, nearest the second end **20** of the support member **16**. As the jack **10** is rotated toward the second orientation **44**, the wider portion of the stabilizing plate **50** at the second end **20** of the support member **16** serves to provide increased stability to the support member **16**, which in this orientation supports the weight of the axle **42** and corresponding vehicle. It will be recognized that the cleats **52** serve to increase traction between the outer arcuate edge of the arcuate edge member **24** and the ground beneath a vehicle as the jack **10** is rotated between the first and second orientations **40**, **44**.

(17) In various embodiments, the jack **10** is provided with a number of stop devices and features in order to limit and/or hinder rotation of the jack **10** beyond the second orientation **44**, and/or to accommodate inadvertent movement of the vehicle beyond the rotational limits of the jack **10**. For example, with reference to FIG. 3, in the illustrated embodiment, the pivot socket **22** of the jack **10** is sized and configured to correspond to a cross-sectional diameter of an axle **42** received therein such that the axle **42** may make contact with both the contact plate **32** and a rear edge **46** of the socket **22** defined by an outer edge of the first bracing member **26**. In certain embodiments, the pivot socket **22** may be sized to correspond to an axle **42** received therein such that the axle **42** may contact both the contact plate **32** and a rear edge **46** of the socket **22**, as well as an inner edge **48** of the first end **36** of the arcuate edge member **24**. In the illustrated embodiment, the first end **36** of the arcuate edge member **24** defines a tapered shape with an inner edge **48** that extends at an outwardly flared angle to the contact plate **32**. Thus, it will be recognized that, in this embodiment, a range of diameters exist of axles which may be received within the pivot socket **22** and which may contact the contact plate **32**, the socket rear edge **46**, and the first end inner edge **48** of the arcuate edge member **24**.

(18) In the illustrated embodiment, the tapered shape of the first end **36** of the arcuate edge member **24** is sized and configured to allow the outer edge **58** of the first end **36** of the arcuate edge member **24** to contact an underside of the vehicle when the jack **10** is rotated to the second orientation **44**. In this regard, in embodiments in which the axle of the vehicle is sized to fit snugly within the pivot socket **22**, with the axle contacting all three surfaces of the contact plate **32**, the rear edge **46** of the socket **22**, and the inner edge **48** of the first end **36** of the arcuate edge member **24**, rotation of the jack **10** toward the second orientation **44** serves to establish a frictional connection, i.e., “wedge,” the first end **36** of the arcuate edge member **24** between the axle and the vehicle. Thus, in various embodiments, the first end **36** of the arcuate edge member **24** serves as a stop to limit rotation of the jack **10** significantly beyond the second orientation **44**.

(19) In certain embodiments, the first end **36** of the arcuate edge member **24** is sized and shaped to accommodate slight rotation of the jack **10** beyond the second orientation **44**. In these embodiments, it will be recognized that over-rotation of the jack **10** beyond the second orientation **44** results in rotation of the support post **16** to a vertical or near-vertical orientation, thus, rotation of the platform **56** beyond the orientation in which the platform **56** lies flush with the ground beneath the vehicle. In other words, over-rotation of the jack **10** beyond the second orientation **44** results in the majority of the platform **56** being lifted from contact with the ground beneath the vehicle and the jack **10** resting on a trailing edge **60** of the arcuate edge member **24**. Thus, in the event the jack is over-rotated beyond the second orientation **44**, the first end **36** of the arcuate edge member **24** may become locked in place between the axle of the vehicle and the portion of the vehicle above the axle. At the same time, the jack **10** may be rotated such that the jack is lifted from the majority of the platform **56** and rests on the trailing edge **60** of the arcuate edge member **24**,

thereby decreasing the surface area of the platform **56** contacting the ground beneath the vehicle and allowing the jack **10** to slide along the ground. In this way, the jack **10** may be maintained beneath the vehicle a short distance as the vehicle is moved beyond the limits of rotation of the jack **10**.

(20) From the foregoing, it will be recognized that a vehicle cam jack **10** is provided which allows for quick, safe, and convenient lifting of an axle portion of a vehicle, while also including additional safety features to accommodate inadvertent over-rotation of the jack **10** beneath the vehicle. In several embodiments, the frame **12** of the jack **10** is of a substantially planar configuration, and the contact plate **32** and stabilizing plate **50** are each of a width less than a few inches, and preferably less than two inches, such that the entire width of the jack **10** is less than two inches. In these embodiments, the jack **10** may be stored in a relatively planar configuration along a vehicle, such as beneath a vehicle, adjacent a side of the vehicle, in a trunk or other storage area of a vehicle, or adjacent a spare tire stored on or in the vehicle.

(21) Numerous variations, modifications, and additional embodiments are possible, and accordingly, all such variations, modifications, and embodiments are to be regarded as being within the spirit and scope of the present general inventive concept. For example, regardless of the content of any portion of this application, unless clearly specified to the contrary, there is no requirement for the inclusion in any claim herein or of any application claiming priority hereto of any particular described or illustrated activity or element, any particular sequence of such activities, or any particular interrelationship of such elements. Moreover, any activity can be repeated, any activity can be performed by multiple entities, and/or any element can be duplicated.

(22) It is noted that the simplified diagrams and drawings included in the present application do not illustrate all the various connections and assemblies of the various components, however, those skilled in the art will understand how to implement such connections and assemblies, based on the illustrated components, figures, and descriptions provided herein, using sound engineering judgment. Numerous variations, modification, and additional embodiments are possible, and, accordingly, all such variations, modifications, and embodiments are to be regarded as being within the spirit and scope of the present general inventive concept.

(23) While the present general inventive concept has been illustrated by description of several example embodiments, and while the illustrative embodiments have been described in detail, it is not the intention of the applicant to restrict or in any way limit the scope of the general inventive concept to such descriptions and illustrations. Instead, the descriptions, drawings, and claims herein are to be regarded as illustrative in nature, and not as restrictive, and additional embodiments will readily appear to those skilled in the art upon reading the above description and drawings. Additional modifications will readily appear to those skilled in the art. Accordingly, departures may be made from such details without departing from the spirit or scope of applicant's general inventive concept.

Claims

1. A jack for lifting an axle portion of a vehicle, the jack comprising: a planar frame comprising a support post and an arcuate edge member; and a pivot socket defined at a first end of the support post, the pivot socket comprising a contact plate positioned at the support post first end and extending perpendicularly outwardly from the planar frame and perpendicular to an elongate dimension of the support post, the pivot socket further comprising a side member defined by the planar frame and extending substantially perpendicular to the contact plate opposite the support post and a tapered member defined by the planar frame and extending substantially perpendicular to the side member and substantially parallel along a planar surface of the contact plate; wherein the pivot socket is sized to receive and engage an axle portion of a vehicle; wherein the arcuate edge member has a first end extending along the tapered member of the pivot socket, a second end

adjacent a second end of the support post, an outer arcuate edge extending outwardly between the first and second ends of the support post, and an offset portion extending outwardly from the second end of the support post opposite the outer arcuate edge; and a stabilizing plate extending perpendicularly from the offset portion and along at least a portion of the outer arcuate edge; wherein the outer arcuate edge defines a radius of curvature greater than a length dimension of the support post; whereby the jack is positionable in a first orientation with the support post extending horizontally and the outer arcuate edge below the support post to allow the pivot socket to receive and engage an axle of a vehicle, and whereby the jack is rotatable along the outer arcuate edge to a second orientation to allow the support post to extend beneath the pivot socket and rest on a portion of the outer arcuate edge adjacent the second end of the support post.

2. The jack of claim 1, wherein the first end of the support jack forms a first side of the pivot socket.
3. The jack of claim 2, wherein the first end of the support jack defines a substantially straight edge.
4. The jack of claim 1 wherein the frame further comprises a plurality of bracing members extending between the support post and the arcuate edge member.
5. The jack of claim 4, wherein a first of the bracing members extends from the first end of the support post to a portion of the arcuate edge member proximate the first end of the arcuate edge member, the first bracing member forming the side member of the pivot socket.
6. The jack of claim 5, the first end of the arcuate edge member forming the tapered member of the pivot socket.
7. The jack of claim 6, the second end of the support post defining a substantially straight edge extending acute to a long dimension of the support post along the offset portion of the arcuate edge.
8. The jack of claim 7, wherein the outer arcuate edge and the stabilizing plate each extend along the substantially straight edge of the support post second end and the offset portion of the arcuate edge to define a support platform.
9. The jack of claim 8, wherein the stabilizing plate further extends along the outer arcuate edge between a midpoint of the outer arcuate edge and the support post second end.
10. The jack of claim 9, wherein the stabilizing plate is tapered outwardly from midpoint of the outer arcuate edge to the support post second end.
11. The jack of claim 10 further including a plurality of cleats defined along the outer arcuate edge at spaced apart locations between the midpoint of the outer arcuate edge and the support post second end.
12. The jack of claim 11, wherein each of the bracing members extends from the support post first end to one of a plurality of spaced apart locations along the arcuate edge member.
13. The jack of claim 12, wherein the support post, bracing members, arcuate edge member, and cleats are integrally formed.
14. The jack of claim 13, wherein the support post, bracing members, arcuate edge member, and cleats are formed from a unitary sheet of rigid material.
